Hedge Funds and Recessions

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Abstract

This paper analyzes the impacts of recessionary periods on hedge funds and mutual funds to understand the differences in their behavior. Through the use of historical returns of three mutual funds, the S&P 500, and the Barclay Index over 26 years, a set of models identifying the effects of the three most recent recessions were constructed. The findings of the models show hedge funds do continue to function without major market influence, which suggests that there are strong diversification benefits to holding hedge funds in a portfolio.

Introduction

Hedge funds are investment pools that, unlike traditional mutual funds, are only available to accredited investors. This has allowed hedge funds to operate under different regulations than those governing publicly traded funds. Critically, hedge funds are not required to publicly report their holdings, and are allowed to hold riskier assets than mutual funds. Hedge funds have historically used these assets to achieve an average alpha of approximately 4% (Stulz, 2007).The investment strategies used to achieve a positive alpha result in hedge funds having a higher expense ratio than mutual funds. In their 2011 study, Ilia D. Dichev, and Gwen Yu found the dollar weighted return of large hedge funds was 3% to 7% lower than that of mutual funds.

Investors still choose to invest in hedge funds for diversification purposes, as hedge funds are less correlated with the market than other investment pools (Amin, Kat, 2003).

To further understand the value and diversification benefit of hedge funds, this paper uses regression models to analyze the difference in the impacts of recessions on hedge funds and mutual funds.

Data

To estimate the difference of the impacts of a recession on hedge funds and mutual funds, the historical data of three mutual funds were compared to the Barclay hedge fund index. The use of a fund of hedge funds allowed for the model to respond to the frequency and effectiveness of the many strategies used to manage these funds. Amin and Kat advise that investors use a hedge fund index rather than investing in individual funds, as individual funds have 5% higher efficiency loss (2003). The Barclay Hedge Fund Index is not a fund of funds; rather, it is an aggregation of the returns of approximately 6,000 hedge funds that aims to represent the behavior of the hedge fund market. The index contains monthly data spanning 26 years, from January 1997 through the end of 2022. This time frame includes three of the most recent recessions, The Early 2000's Recession, The Great Recession, and the Covid recession; the periods of recession indicated by the Saint Louis FED were used. Unfortunately for the analysis, the Bureau of Economic Analysis attributes these recessions to different causes and has found many differences in their impacts (2018, 2021). This restricted the effectiveness of comparing the impacts of one recession to another. However because funds were exposed to the same conditions under each recession, the comparisons made between different funds during the same recession period are valid.

Due to mutual funds requirements to publish their financial records to the public a greater number of selection factors were considered. Any fund that was not actively traded from 1997 through 2023 was not considered to avoid further restricting the sample size. This analysis was not done to find an investment firm or asset class that are resilient to recessions, therefore to avoid bias from confounding variables, mutual funds offered by multiple firms, with different primary asset types. Finally, to mirror the management style of hedge funds, the mutual funds considered had very active management styles. To consider this criteria the mutual funds with high expense ratios for each firm were considered, as the time and other resources put into managing a fund will be reflected in the expense ratio. A set of three funds were determined to adequately meet this criteria, Vanguard Dividend Growth Fund (VDIGX), Schwab Core Equity Fund (PGSGX), Fidelity Value Fund, (FDVLX). The S&P 500 was used as a proxy for the U.S. stock market. The forces acting on the market are captured well by the S&P 500's returns because of its wide diversification and high volume of trading.

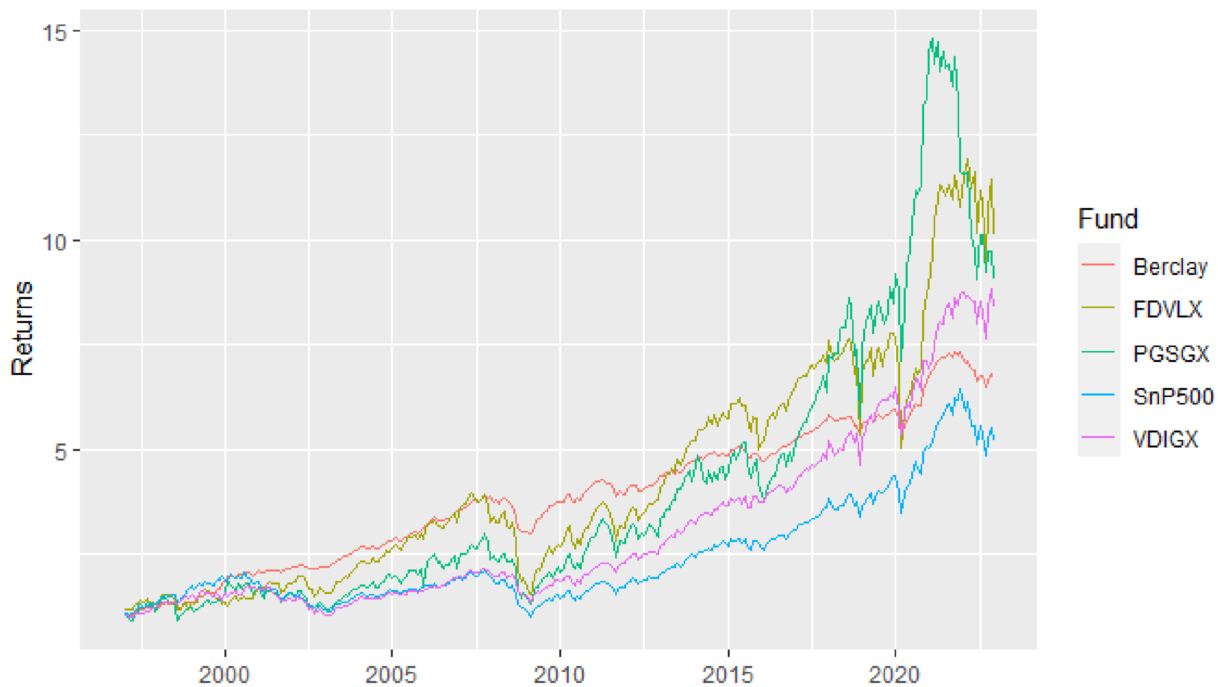
Methods

Table 1

Variables	Barclay	VDIGX	FDVLX	PGSGX	S&P 500
Mean	0.006119	0.0068459	0.0070958	0.0074413	0.0052878
Variance	0.0004284	0.001572	0.0053537	0.0041667	0.0020542
Sharpe Ratio	0.2956308	0.1726638	0.0969791	0.1152796	0.1166697
Holding Period Return	5.7081322	7.403406	8.0801128	9.1055533	4.1833302

Initially, each fund was summarized by basic statistical analysis, shown in Table 1. The geometric mean of returns for the Barclay Index is lower than that of all three mutual funds. This suggested that the landscape of funds observed in Dichev and Yu 2011 still holds. This is further confirmed by the holding period returns, where the Barclay Index was again ranked lower than all the mutual funds. The percent difference between the mean return of the Barclay Index and the ETFs were higher than the percentage difference in HPR. For VDIGX, the fund with the closest mean and HPR, these statistics were 11.87% and 22.89% higher respectively. This pattern is explained by the comparatively very low variance of the Barclay Index to the mutual funds; the variance of VDIGX was over 250% higher. The HPR was calculated by hypothetically investing one dollar into each of the funds at the start of January 1997 and holding the asset to the start of 2023. The returns for each of the funds is shown as a time series in Figure 1.

Figure 1



This plot highlighted many key characteristics of the dataset. The Great Recession and the Covid Recession both show a sizable downturn in all of the funds as well as the overall market, however the Early 2000's Recession does not show the same deviation from the standard trend. Immediately following Covid Recession, each fund and index grew at their highest rate, followed by a deflation trending back to values more in line with the overall trends. Overall, while the Barclay Index reacted to the recessionary periods, their impact was much less pronounced, indicating that hedge funds do react differently to the stress of a recession.

The difference in behavior in the time series of returns prompted the use of an Analysis of Variance (ANOVA) test to further establish the differences present in groups.

Figure 2

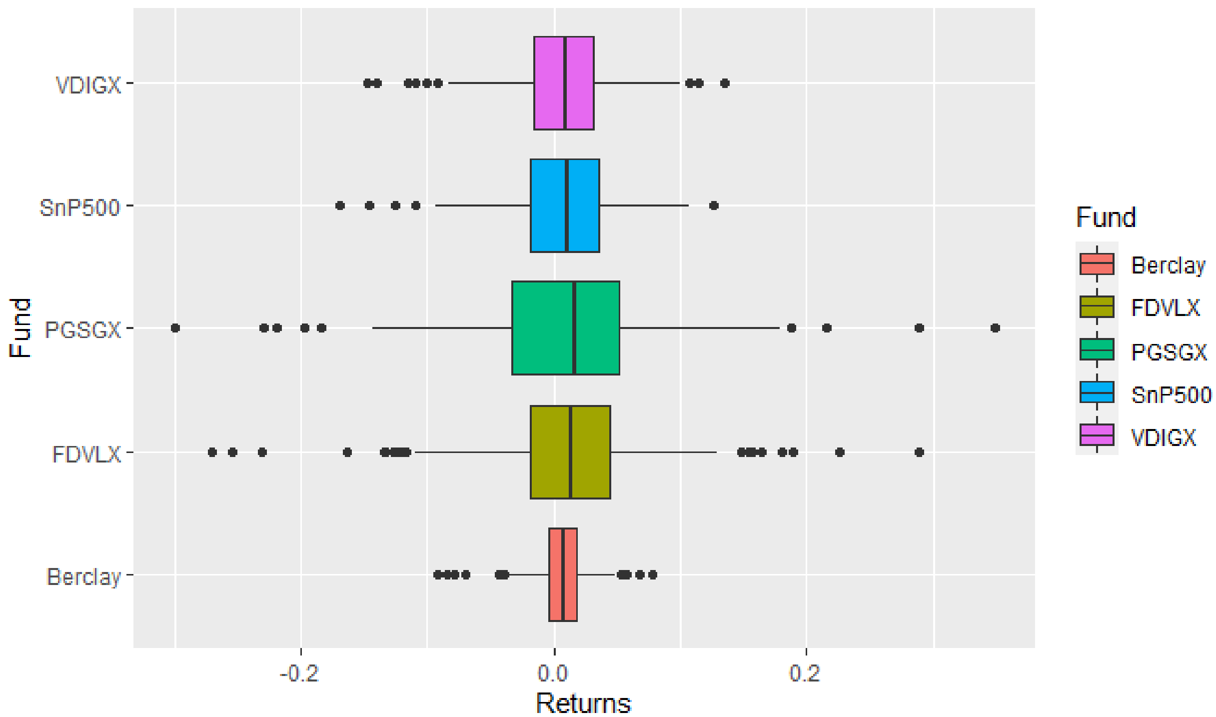


Figure 2 depicts the test results through the utilization of a boxplot, which showcases the distribution of returns among the various funds. The median monthly returns appear to be statistically and visually similar, as evidenced by a p-value of .862. Thus, the focus of the analysis was shifted to the variance shown by the plot. All but one datum from the Barclay Index's returns fall within the fenced range of any mutual fund, they would not be considered outliers under the variance of any other fund.

The correlation between the differences in the variance of returns and recessions was further explored through the construction of a set of linear models. The models used the excess return of the market and indicator variables for the periods of recession along with the interaction term of these two variables. The historical returns for the S&P 500 were used as a proxy for the market's return. A Dicky Fuller test confirmed that each series of returns was a stationary process. The constructed models underwent a formal test for heteroscedasticity using the Breusch-Pagan method, which yielded no significant evidence of heteroscedasticity. Table 2 shows the coefficients for each term in the linear model.

Table 2

Variables	Barclay	VDIGX	FDVLX	PGSGX
Market Return	0.3730744	0.6943061	1.0250035	1.1266078
Great Recession	0.0021289	0.0043874	0.0139275	0.0152488
2000's Recession	0.005857	-0.0155545	0.0079044	0.0070433
Covid Recession	0.1012252	-0.0472934	0.243122	0.2854382
Market X Great Recession	0.027194	0.0869656	0.3910944	0.0529858
Market X 2000's Recession	-0.1310076	-0.061408	-0.1571014	-0.0475028
Market X Covid Recession	1.1680562	-0.2667936	3.0779912	2.7258287

Each model showed a significant, positive correlation, at the 5% level, between an increase in market return and an increase in the assets return. PGSGX and FDVLX models both have a coefficient of over one for the market's return, therefore these fund's returns move in the direction of the market with greater magnitude. Whereas, the Barclay Index model indicated that for every 1% change in returns for the market, hedge funds only experience a .37% shift. The pure recession coefficients represent a static change in the expected returns of the fund for the period of recession.VDIGX was the only fund that had any significant changes to their estimated return through the Early 2000's Recession, at the 10% level. The Great recession did not have a significant flat impact on the return of any of the funds. The Covid Recession, at a 5% level, significantly increased the returns of the Barclay Index by 10%. The interaction variables show how each recession altered the relationship between the return of the market and the fund's returns. The Covid Recession caused significant shifts to the relationships of the Barclay Index and FDVLX, at the 5% level, with coefficients of 1.17 and 3.08 respectively. The ratio of FDVLX and the Barclay Index's coefficients during the Covid recession was 2.75, the ratio outside of the Covid Recession was 2.64. This implies that FDVLX held about 2.7 times more assets dependent on the market compared to Barclay.

The behavior of the Barclay Index compared to the mutual funds, both in the regression models and the other statistical analysis, highlights the difference in hedge funds and mutual funds as investments. Hedge funds are often thought of as entities that use their lack of regulation to take on risk. However, the calculated low dependence on market performance and historically low variance of hedge funds provides evidence that the average hedge fund does not engage in risky behavior. Selling derivatives or shorting stock has not forced hedge funds to take on an excessive amount of risk, which shows they have hedged away large amounts of their risk. With strategies such as delta hedging, a large amount of capital under constant management can be used to isolate small profits. In addition, the significant positive impact of the pure Covid Recession coefficient can be explained by these ideas. A properly hedged position has the same expected value regardless of the movement in prices of the underlying asset.

The models were not flawless, the largest issue was the availability of data. Hedge funds exercise their rights to privacy, and do not disclose their historical returns, restricting analysis to the Barclay Index. The index has 26 years of data, not immediate cause for concern, however that timespan only contained three recessions. To preserve the few recessions in the dataset, only funds with trading data from before 1997 were considered. This further restricted available data, as well as likely introduced some level of selection bias into the models. Regardless of these flaws, the model still shows statistically significant trends that are logically sound.

Conclusion

Through the use of the linear regression models, the behavior of hedge funds contrasted to mutual funds, particularly through recessionary periods. With a low correlation to the market and a fixed rate of return through the Covid recession, hedge funds provide a valuable diversification benefit to the portfolios of those with opportunities to invest in them.

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